Assessment 1

DDD 2: Introduction to Programming and Basic Data Analysis

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Question 1

```
x <- c(4, 7, 2, 9, 5, 6)
x > mean(x)
```

```
[1] FALSE  TRUE FALSE  TRUE FALSE  TRUE
```

```
#The output TRUE indicates elements greater than the mean of the vector.
```

Question 2

```
df <- data.frame(
  id = 1:4,
  score = c(65, 70, 68, 72),
  group = c("A", "B", "A", "B"))
df[df$id == 2, "score"] <- 75
df # again to check new updated one variable in our data frame
```

```
  id score group
1  1    65     A
2  2    75     B
3  3    68     A
4  4    72     B
```

Question 3

```
x <- c(1, 2, 3, 4)
y <- x
y[2] <- 10 #MODIFY ONE ELEMENT OF Y VARIABLE

#CHECK
x#x does NOT change
```

```
[1] 1 2 3 4
```

```
y#y changes
```

```
[1]  1 10  3  4
```

Question 4

```
df <- data.frame(
  x = c(2, 4, 6, 8, 10, 12, 14, 16),
  y = c(5, 7, 9, 11, 13, 15, 17, 19))
df[] <- lapply(df, function(z) z - mean(z))
df
```

```
   x  y
1 -7 -7
2 -5 -5
3 -3 -3
4 -1 -1
5  1  1
6  3  3
7  5  5
8  7  7
```

Question 5

```
v <- c("Low", "Medium", "High", "Medium", "Low", "High")
f <- factor(v, levels = c("Low", "Medium", "High", "VeryHigh"), ordered = TRUE)
f
```

```
[1] Low    Medium High   Medium Low    High
Levels: Low < Medium < High < VeryHigh
```

```
as.numeric(f)
```

```
[1] 1 2 3 2 1 3
```

```
typeof(f)
```

```
[1] "integer"
```

Question 6

```
mean_na <- function(x) {
  mean(x, na.rm = TRUE)}
#to ceeck it we are taking an example
v <- c(2, 4, NA, 6, 8)
mean_na(v)
```

```
[1] 5
```

Question 7

```
# Create data frame with 6 observations
df <- data.frame(
  id = 1:6,
  age = c(22, 27, 24, 30, 26, 21),
  gender = c("F", "M", "F", "M", "F", "M"),
  income = c(20000, 35000, 25000, 40000, 32000, 18000))
#(a) Rename variables
colnames(df)[colnames(df) == "age"] <- "age_years"
colnames(df)[colnames(df) == "income"] <- "annual_income"
#(b) Add indicator variable (age > 25)
```

```r
df$older_than_25 <- df$age_years > 25
#(c) Subset individuals older than 25
df_sub <- df[df$older_than_25 == TRUE, ]
#(d) Keep only id and indicator variable
final_df <- df_sub[, c("id", "older_than_25")]
final_df
```

```
  id older_than_25
2  2          TRUE
4  4          TRUE
5  5          TRUE
```

Question 8

```r
#Create data frame with four numeric variables
df <- data.frame(
  a = c(2, 4, 6, NA, 8),
  b = c(1, 3, 5, 7, NA),
  c = c(10, 12, NA, 14, 16),
  d = c(5, NA, 9, 11, 13))
#Define function
median_dev_fun <- function(data) {

  data$a_md <- data$a - median(data$a, na.rm = TRUE)
  data$b_md <- data$b - median(data$b, na.rm = TRUE)
  data$c_md <- data$c - median(data$c, na.rm = TRUE)
  data$d_md <- data$d - median(data$d, na.rm = TRUE)

  data$sum_md <- data$a_md + data$b_md + data$c_md + data$d_md

  data
}

#Apply function
result <- median_dev_fun(df)
result
```

```
   a  b  c  d a_md b_md c_md d_md sum_md
1  2  1 10  5   -3   -3   -3   -5    -14
2  4  3 12 NA   -1   -1   -1   NA     NA
3  6  5 NA  9    1    1   NA   -1     NA
```

```
4 NA  7 14 11   NA    3    1    1     NA
5  8 NA 16 13    3   NA    3    3     NA
```